

Modulus-Gradient Conductive Core–Shell Structures Formed by Magnetic Self-Assembling and Printing Processes for Highly Stretchable Via Applications

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In recent years, stretchable electronics have been widely studied in multidisciplinary fields of science and engineering, offering the attractive prospect of linking material science, chemistry, physics, and electronics. Starting from improving performance of a single stretchable electrode and/or a device unit, such as a transistor or a sensor, significant efforts have been made to implement intimately connected multifunctional systems for many practical applications, such as ubiquitous healthcare,^[1–3] soft robotics,^[4–6] or novel user interfaces.^[7] Existing techniques, however, have mostly adopted a low level of system integration and functional density in conjunction with sophisticated fabrication processes and circuit configuration, due to spatial limitation of 2D planar substrate.

To meet the criteria required for the aforementioned high-density system, a “via”—also termed as a “Vertical Interconnect Access”—is thought to be the most promising pathway. In general, via technology is being used widely to make a conductive path between layers of boards in the case of conventional printed circuit boards (PCBs). By penetrating the boards and filling the holes with electrically conductive materials, the upper and lower surfaces of the PCBs are electrically connected. Different from the conventional rigid board, however, such a conventional drilling/filling process is an onerous task for soft stretchable systems. Although a few studies have reported conductive vias for stretchable substrates in the form of liquid metal,^[8,9] and solid-phase materials,^[10–12] obvious limitations have remained. In the case of liquid metal, which is the most widely used as a highly conductive stretchable conductor, channel formation and filling processes through a stretchable substrate are necessarily required. Moreover, due to the liquid-phase property, it is difficult to encapsulate a physically opened via hole and form an electrical contact between the filling materials and other circuitry components simultaneously.^[13] In the case of solid-phase stretchable vias, additional drilling/filling process restricts processability. Besides, the reliability under mechanical deformation of vias was rarely investigated (Table S1, Supporting Information). As a result, there is a

rapidly increasing need to investigate materials, structural design, and integration tools of mechanically robust vertical interconnects for stretchable electronic systems.

Here, we report a magnetically self-constructed, mechanically robust, and electrically conductive core–shell structure that is site selectively patterned and monolithically embedded into the soft substrate itself. The conductive core–shell structure plays an excellent role of the predefined vertical interconnect (via) in a stretchable form. In combination with strongly focused magnetic field and site-selective printing process, the mixture of nickel (Ni) microparticles and silicone resin was spontaneously formed into the core–shell structure by the magnetic-field-induced self-assembly process. This modulus-gradient structure prevents the core–shell via from failure that can be induced by mechanical deformation, and helps to maintain its own electrical property even under a harsh strain condition. Digital image correlation (DIC) analysis on strain distribution and measurement of several electrical properties confirmed the mechanical and electrical reliability of our core–shell via structures. Finally, in order to show the feasibility of our approach in the stretchable electronics area, a stretchable, multilayered pressure sensing/visualizing system was successfully demonstrated.

Figure 1a briefly shows schematic drawings of the fabrication process of a core–shell via embedded in a stretchable substrate. First, Ni/silicone resin mixture was prepared at 1:1 weight ratio of Ni micro-particles (<50 μm) and OE-6630 silicone resin. The mixture was dispensed onto the glass substrate by liquid dispensing system, followed by pouring uncured substrate material (poly(dimethyl siloxane), PDMS) just onto the mixture dispensed glass. Polyimide film (Kapton tape) spacers were placed at edges in order to obtain about 300 μm thick stretchable PDMS substrate. After the subsequent vacuum degassing and PET plastic film covering, curing process was performed. Concurrently, a pair of magnetized iron pillar arrays was placed and aligned right above and under the positions of each via so that it generates selective magnetic field just right through the via. Strongly focused magnetic field quickly assembled Ni particles to the center of the via during the curing process, forming a conductive Ni core surrounded by a residual silicone resin that served as a rigid shell. This rigid shell prevents the Ni core from mechanical deformation, showing high mechanical and electrical stability under the tensile strain (detailed fabrication process and mold setting are described in the Experimental section and Figures S1 and S2 in the Supporting Information).

For characterization and process optimization of a core–shell via, the effect of magnetic flux density on electrical

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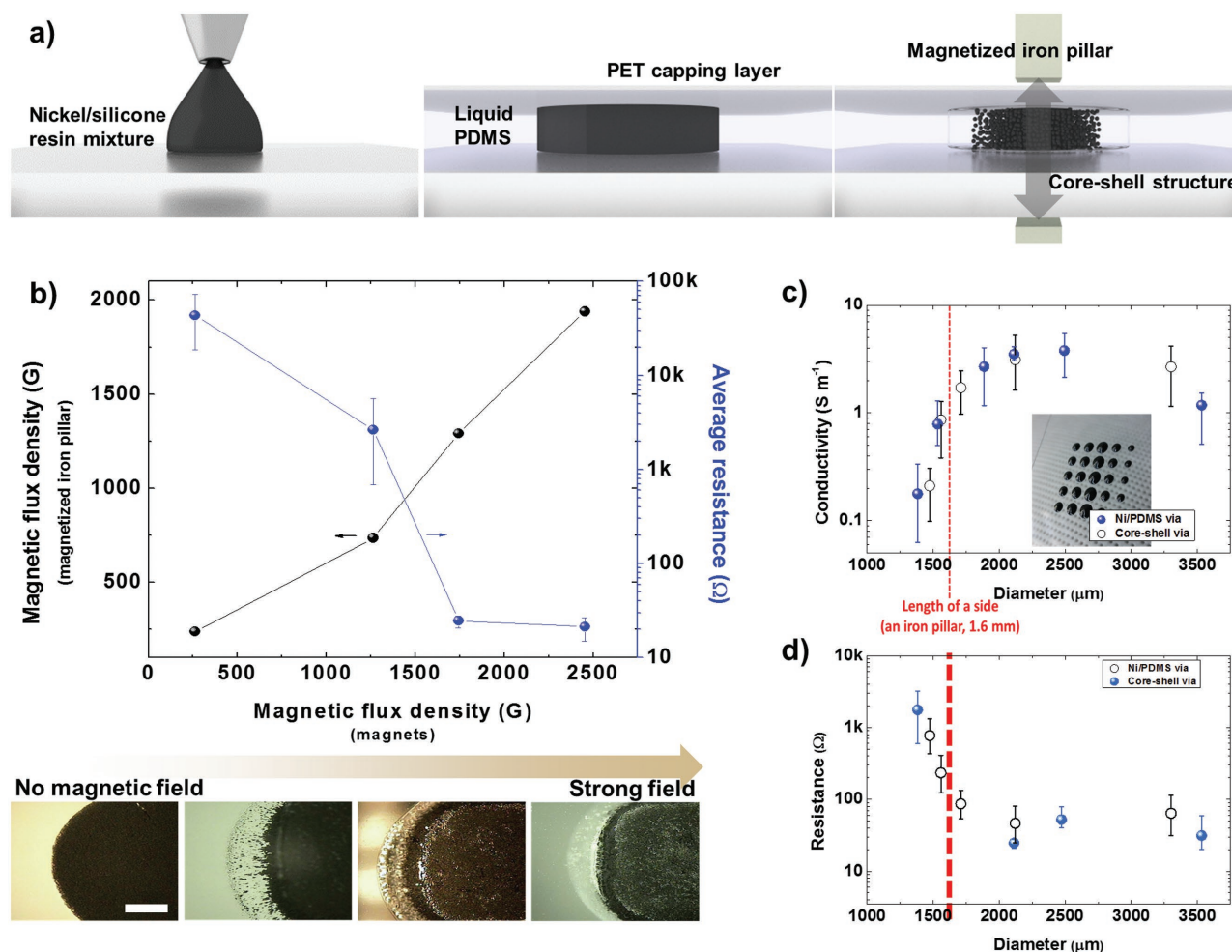


Figure 1. Fabrication process and optimization of a core-shell structure. a) Schematic drawings of the fabrication process. b) Resistance of the magnetically aligned vias corresponding to the strength of magnets. Optical images below the graph show morphological changes of dispensed Ni/silicone drop according to the magnetic flux density. Scale bar is 500 μm . c) Conductivity and d) resistance of core-shell structure according to the diameter.

characteristics of vias was investigated. Same amount of mixture was dispensed for all via-fabrication process. Gradual magnetic flux density (approximately from 200 to 2000 G) was obtained by several species of commercial magnets, e.g., ferrite, samarium-cobalt (SmCo) and neodymium (Nd)-based magnets. Eutectic gallium indium alloy (495425; Sigma-Aldrich) was used as an electrical contact material for all measurement. As shown in Figure 1b, Ni particles centrally migrated in the mixture drop on the existence of magnetic field. The stronger the magnets, the more nickel particles could be gathered to the center. It is noted that the resistance of vias was quite high when weak magnets were used, probably due to loose packing of Ni particle. On the contrary, in the case of high magnetic flux density strength over 1700 G, which is obtained by SmCo and Nd magnets, Ni particles congregated densely above the percolation threshold, resulting in sufficiently low resistance ($<30 \Omega$). In our experiment, SmCo-based magnets were used to magnetize iron pillar arrays because Nd-based magnets easily lose their magnetic property under high temperature. Next, we varied the size of vias and analyzed the electrical properties of each via (Figure 1c,d). The diameter of each via was

controlled by adjusting the air pressure and dispensing time of the dispenser. As expected, the conductivity of vias drastically increased when the size of vias was enlarged. But when the diameter of vias was bigger than the side length of the iron pillar (the right hand side to the dotted red line in Figure 1c,d), it seemed that surplus Ni particles did not further contribute to decrease of via resistance, probably due to the nonalignment of the off-centered particles and limit in the concentration of the Ni particles moving to the center.

We also analyzed the morphological characteristics of a core shell via by optical images and SEM observation. Figure 2a shows well-defined core-shell vias embedded in the PDMS substrate by dispensing mixture followed by the magnetic alignment. A via-embedded substrate was stretchable and conformable to arbitrarily curved surface (right images of Figure 2a). More detailed structure was displayed in Figure 2b. A core-shell structure has three compartmental regions each having different elastic modulus (Figure S3, Supporting Information). This modulus gradient suppresses separation between Ni composite and substrate even for their large modulus difference.^[14] To further investigate the microscale morphology of a via, field

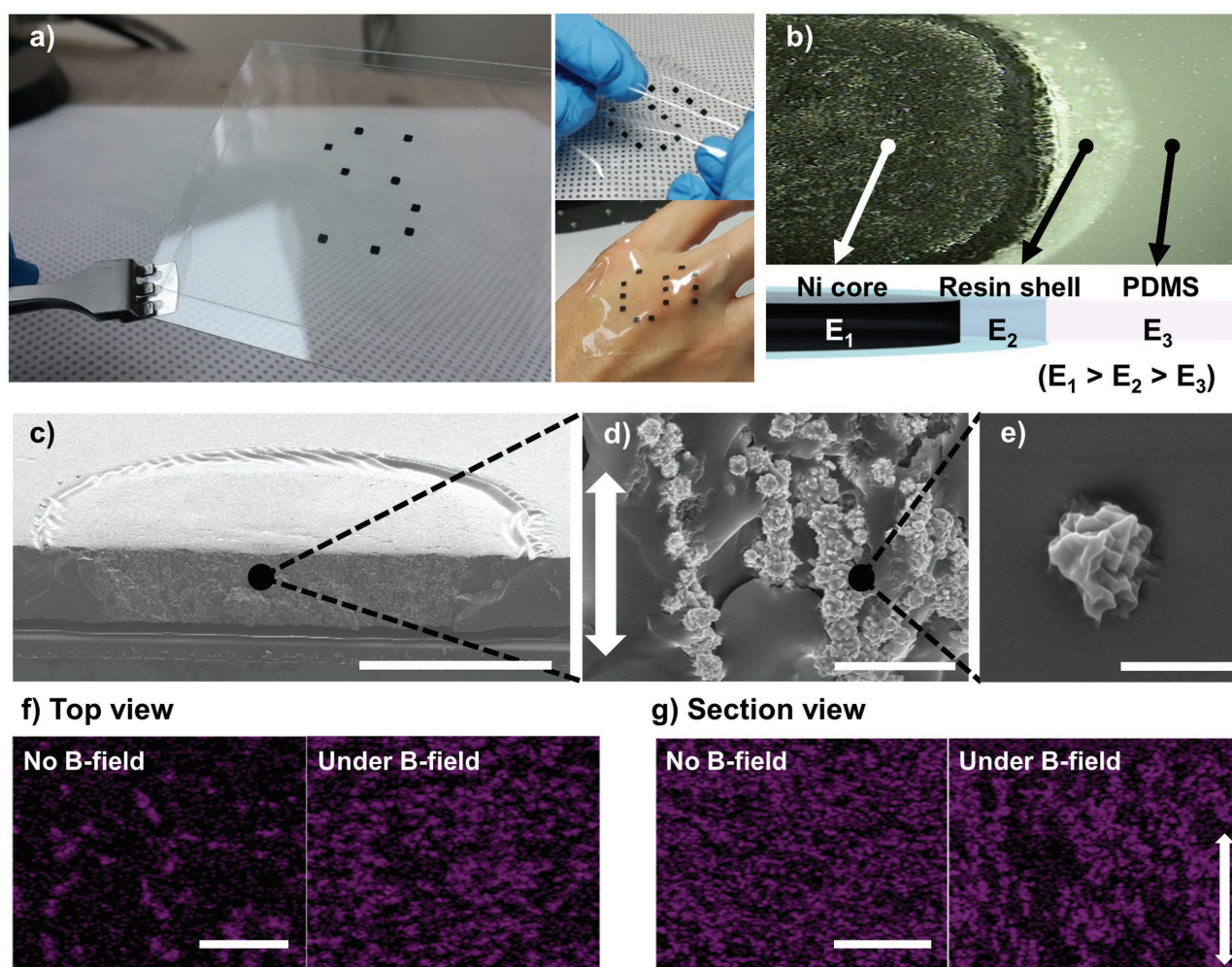


Figure 2. Morphology of core-shell vias. a) An example of patterned core-shell vias embedded in the PDMS substrate. b) A schematic model of a core-shell via with compartmental structure. c–e) SEM images of a sliced core-shell via and vertically aligned nickel particles inside polymer matrix. Double-sided arrow bar indicates the direction of applied magnetic field. Scale bars: c) 1 mm, d) 10 μm , and e) 1 μm . f) Energy-dispersive spectroscopy (EDS) images of top view and g) section view of the randomly distributed via and magnetically aligned core-shell via. Scale bars are 25 μm in top view and 100 μm in section view.

emission scanning electron microscope (FE-SEM) images were taken (FE-SEM II S-480; Hitachi) with different magnifications (Figure 2c–e). From Figure 2d, obvious vertical alignment of Ni particles along the direction of the applied magnetic field was observed. This anisotropic arrangement of Ni particles inside the via can lower the vertical resistance of composite.^[15–17] Also, irregular shape of Ni particles is thought to make strong electric field at the sharp edges of each particle under electrical bias condition, causing efficient tunneling conduction through the dielectric matrix between Ni particles (Figure 2e).^[18] The spatial distribution of the Ni element in the via was also mapped by using energy-dispersive X-ray spectroscopy (EDS) in field emission scanning electron microscope (FE-SEM 7800F Prime; JEOL Ltd.) to precisely observe the behavior of Ni particles (Figure 2f,g). Each purple-colored dot indicates positions of Ni. As predicted, nickel particles were buried under the superficial resin and randomly distributed in case of zero magnetic field. By contrast, it seems that strong magnetic field induced penetration and arrangement of Ni particles inside the resin.

In order to confirm mechanical and electrical reliability of a core-shell structured via under mechanical deformation, several experiments were performed. First, the strain distribution on and around the via was analyzed by DIC (VIC-2D; Correlated Solutions) for the given tensile strain.^[19,20] For the exact calculation of local strain of elastomeric materials, Hencky strain tensor was used as a strain tensor.^[21–24] As a comparative group, characteristics of a Ni/PDMS via, which has also aligned by magnetic field but has no modulus-gradient property, was also examined in parallel under the same conditions. Figure 3a,c shows the color mapping of strain distribution on top view and section view of vias under stretch. In all cases, average strain values of a core-shell via remained the lowest at 50% strain. These results indicate that the volume change of core-shell structure was insignificant compared to that of Ni/PDMS via, so that the aligned nickel particles were not agitated. Precise strain values are plotted in Figure 3b,d. It is notable that the values of strain along the y -direction (ϵ_{yy}) were negative due to the Poisson's effect.^[25] This protection effect of the core-shell

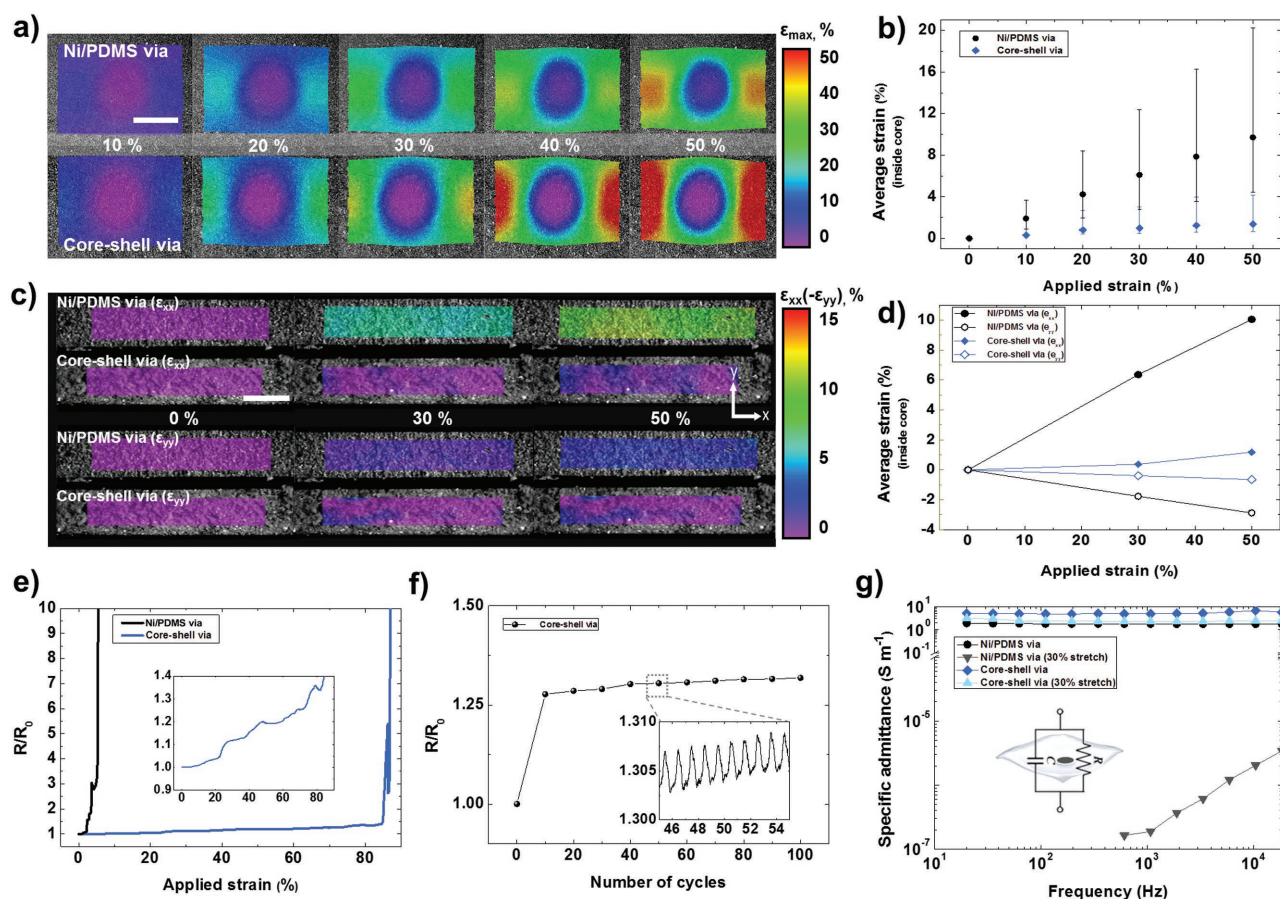


Figure 3. Mechanical and electrical characterization of core-shell vias. Strain distribution of a,b) the top view and c,d) the section view for core-shell vias and Ni/PDMS vias under uniaxial stretching. $\epsilon_{\max}$ was the largest value among principal strain values, ϵ_{xx} and ϵ_{yy} are strain values along each axis. Scale bars: a) 2 mm, c) 300 μm . e) Plot of resistance change versus applied strain for vias. f) Cyclic test of the core-shell via at 30% tensile strain. g) Impedance spectroscopy of a core-shell via and a Ni/PDMS via.

structure was also available under two dimensionally applied strain, sustaining internal strain ($\epsilon_{\max}$) of the core under 3% at 25% strain (Figure S4, Supporting Information).

For electrical characterization, we used vias with 2.1 mm diameter embedded in the PDMS substrate with 2 cm in length and 1 cm in width. Figure 3e presents a plot of normalized resistance change versus applied strain up to 100%. Stretching speed was sustained at 1 mm min⁻¹. During stretching, the core-shell via showed a very low level of normalized resistance change under 1.4 even at $\approx 80\%$ strain, which is superior to that of Ni/PDMS via. We also conducted cyclic test of tensile loading (Figure 3f) of 30% strain at 100 mm min⁻¹. Until 100 cycles, no notable resistance change was found except initial five cycles, showing the 25% increased resistance compared to the initial resistance. This initial resistance change is typically observed in a composite-based conductor and is thought to be mainly attributed to the initial displacement of the aligned Ni particles. As shown in the inset of Figure 3f, resistance change in one cycle was less than 1% at the 50th cycle. A cycle test under more severe conditions was also carried out (stretching speed: 500 mm min⁻¹, applied strain: 50%) and showed no remarkable resistance change during the test (Figure S5, Supporting Information).

Since, as previously reported,^[24] impedance spectroscopy (IS) well describes the behavior change of the conductive composite under the tensile strain condition, we utilized IS for further analysis of electrical performance change. A via could be modeled as parallel connection of a resistor (R) and a capacitor (C) (inset schematic in Figure 3g) since it consists of conductive path along the Ni particle and silicone resin dielectric. Therefore, complex admittance (Y^*) is determined as a function of frequency (ω) described as^[26,27]

$$Y^*(\omega) = \frac{1}{R} + j\omega C \quad (1)$$

Considering the dimension of a via, specific admittance is given by

$$y(\omega) = |Y^*| \frac{d}{A} \quad (2)$$

where d is thickness of the via and A is contact area of the via. Specific admittance to the frequency for each via is plotted in Figure 3g. In the case of core-shell structure, specific admittance remained nearly constant regardless of frequency change and tensile deformation, showing mainly ohmic property. On the other hand, in the case of Ni/PDMS, specific admittance

tended to increase as frequency increase when 30% tensile strain was applied. It is in agreement with previous results that a via with no rigid shell suffers a severe volume change by the external force, promoting misalignment and contact failure between Ni particles. As a result, the ohmic component disappears and the capacitive component affects the specific admittance ($Y^* \approx j\omega C$) predominantly, implying dielectric properties of the stretched Ni/PDMS via.

Due to the novel monolithic via-embedded structure, a stretchable multistacked electronic system can be implemented by a simple laminating process. To exhibit its practical applicability, we demonstrated an integrated system composed of a display layer and a pressure-sensing layer. First, a simple circuit composed of surface-mount-device light emitting diode

(SMD LED) chip and inkjet-printed buckled interconnects was implemented (details of fabrication process are described in the Experimental Section). It is worth noticing that a monolithically prepared via-embedded substrate is simply adapted into a stretchable platform by prestretching it and direct printing onto it.^[28–31] Figure 4a shows I – V characteristics of an implemented SMD LED chip in conjunction with a core-shell via. Despite slightly decreased current level due to resistance change of the via, it is enough to light an LED chip even under 2D strain ($\approx 19\%$ change in resistance at $\epsilon_x, \epsilon_y \approx 20\%$). After the LED array was fabricated, it was laminated on a piezoresistive sensing sheet. The schematic and optical images of the layered system are shown in Figure 4b,c, respectively. The two layers were chemically bonded with oxygen-plasma bonding technique

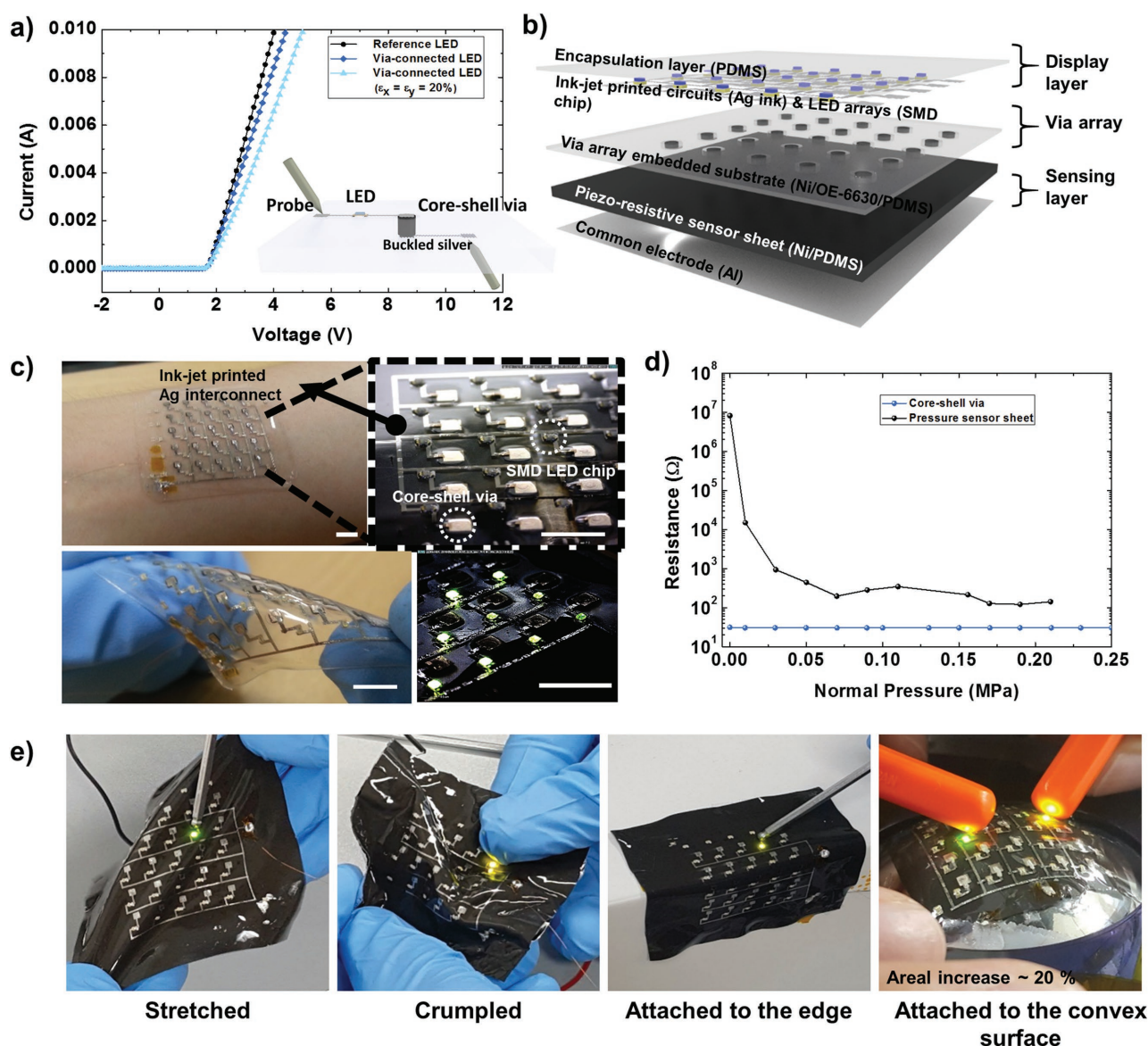


Figure 4. Application of core-shell vias. a) I – V characteristic of via-applied SMD LED chips. b) Schematic of the pressure visualization system as a multilayered system application. c) Optical images of the display layer with inkjet-printed interconnects, SMD LED chips, and embedded core-shell via array. Scale bars are each 5 mm. d) Plot of resistance versus normal pressure of a core-shell via and underlying pressure sensor sheet. e) Optical images of the pressure visualization circuit operation under various situations.

(details in the Experimental section).^[32] When pressure is applied, the system shows intensity and positions of the applied normal pressure with light. When normal pressure is applied to a certain position, resistance of the compressed area of sensor sheet under the core-shell via is drastically decreased, causing a current flow at that location.^[17,33,34] However, the core-shell via is pressure insensitive, due to its mechanical rigidity, hence it can sustain its low resistance as a vertical interconnect under the applied pressure (Figure 4d). By this way, the information of applied pressure can be transduced into positions and intensity of light from the LED array. Furthermore, our system is capable of operating in various deformation conditions such as stretching or crumpling, without remarkable performance degradation (Figure 4e; Figure S6, Supporting Information).

In summary, we demonstrated a magnetically self-assembled, modulus-gradient conductive core-shell structure for the stretchable vertical interconnect. Patterned strong magnetic field concentrates the nickel particles toward the center and core-shell structures were formed spontaneously inside the stretchable substrate during the process of substrate solidification. By this way, mechanical robustness of the via was greatly enhanced and it was able to endure large stress by its novel outwardly modulus-gradient structural property. Additionally, since our core-shell structures can be monolithically embedded in situ during formation process of the stretchable substrate, they are compatible with other types of elastomeric substrate materials and there is no requirement of any special strategy such as drilling and filling process, compared to other stretchable via-fabrication process. Furthermore, demonstration of via-embedded LED-array layer laminated on the pressure sensor sheet provided a promising pathway toward layer-by-layer multiple stacking of so-called "stretchable printed circuit board" for the highly integrated stretchable electronic system applications. Relatively low conductivity in comparison with the conventional vias can be further improved by using nickel composite materials with 1D structure or introducing other ferromagnetic materials with higher electrical conductivity.

Experimental Section

Fabrication of Core-Shell Structures: Nickel particles (26 6981; <50 μm , 99.7%, Sigma Aldrich) were mixed into the thermally curable silicone resin (OE-6630 A and B; mixing ratio of 1:4, Dow Corning Corp.). The weight percent of nickel powder was 50 wt%. The mixture was directly dispensed onto the water-soluble sacrificial layer (poly(vinyl alcohol), PVA) (36 3073; $M_w = 31\,000$ – $50\,000$, 87%–89% hydrolyzed, Sigma Aldrich) coated glass by a fluid dispensing system (SHOT mini 200 SX; Musashi Engineering). Sacrificial layer was prepared with 5 wt% of PVA aqueous solution by spin-coating onto the glass with spin speed 1000 rpm for 1 min. Poly(dimethyl siloxane) (Sylgard 184; mixing ratio of 20:1, Dow Corning Corp.) was poured onto the mixture dispensed glass and covered by plastic thin film (poly(ethylene terephthalate), PET) with certain gap ($\approx 300\,\mu\text{m}$) maintained by polyimide spacers. The stepped curing process was conducted from 60 $^{\circ}\text{C}$ to 150 $^{\circ}\text{C}$ for 6 h. During the curing process, a pair of magnetized iron pillar arrays with high aspect ratio (≈ 3.75) applied selective magnetic field onto the dispensed mixture. After the curing process, PVA sacrificial layer was dissolved by immersing it into deionized water with ultrasonication for 30 min.

Fabrication of Inkjet-Printed Circuits: A PDMS substrate with core-shell vias was prestretched about 20% in two perpendicular directions, followed by surface treatment with UV-ozone cleaner (AH1700;

AHTECH LTS) for 25 min to make the surface more hydrophilic. Silver nanoparticle ink (DGP-40; ANP Corp.) was directly printed onto the surface-treated PDMS by inkjet printing system (DMP-2831; Dimatix Corp.) with specific circuit configuration. Sintering process was followed at 125 $^{\circ}\text{C}$ for 30 min. To place and solder the SMD chips, silver epoxy (ABLEBOND 84-1LMISR4; Henkel) was dispensed onto the inkjet-printed pad as soldering material by a dispenser. Each SMD chip was then placed by pick-and-place machine (TM220A; Hangzhou NeoDen Technology Co., Ltd.) at accurate positions. Sintering process was followed at 175 $^{\circ}\text{C}$ for 1 h. Prestretched substrate was then released to induce wrinkles on the printed silver interconnects.

Fabrication of a Pressure-Sensitive Sheet: Ni particles and liquid PDMS (mixing ratio of 20:1) were mixed at a 1:1 weight ratio and spin-coated onto the glass with spin speed 600 rpm for 1 min. Spin-coated Ni/PDMS mixture was cured at 125 $^{\circ}\text{C}$ for 30 min without external magnetic field to obtain high initial resistance and reasonable sensory characteristic.

Bonding Process of Two PDMS-Based Functional Layers: A display layer and a pressure sensor sheet were surface treated by oxygen plasma system (CUTE-MPR; Femto Science Inc.). Applied power was 48 W and plasma generation time was 42 s. During the process, atmospheric pressure was maintained at 0.5 torr. Surface-treated two interfaces were knitted together and heated at 40 $^{\circ}\text{C}$ for 10 min.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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